

Topology-Aware & Hierarchical All-Reduce; Libraries

- Libraries:
 - NCCL (GPU-centric), Gloo (CPU/GPU), MPI (HPC)
- Algorithms:
 - Ring: bandwidth-optimal for large messages
 - Tree: lower latency for smaller messages
 - Reduce-scatter + all-gather: core of sharded training
- Hierarchical all-reduce:
 - Intra-node reduce-scatter (NVLink)
 - Inter-node reduce-scatter (IB/Ethernet)
 - Inter-node all-gather
 - Intra-node all-gather
- Practice: bucket sizes, algorithm selection per tensor size, topology hints